pharmaniaga®

Cephalexin

Capsule

DESCRIPTION

Pharmaniaga Cephalexin Capsule 250 mg.
White granules in pink / pink hard gelatin capsule size '2' with Pharmaniaga icon and '250' markings.

Pharmaniaga Cephalexin Capsule 500 mg.
Clean, dry, white / pink hard gelatin capsule size '0' with Pharmaniaga icon and '500' markings.

COMPOSITION

Each capsule contains Cephalexin equivalent to 250 mg or 500 mg of Cephalexin anhydrous.

PHARMACODYNAMICS

Cephalexin is an oral broad-spectrum antibiotic belonging to the group of semi-synthetic cephalosporins. In adequate concentrations it is bactericidal for sensitive proliferating micro-organisms by inhibiting the biosynthesis of the cell wall. Even patients with sensitive stomach tolerate the drug well.

Spectrum of action

Of the pathogens listed below, most strains show sensitivity.

Gram-positive organisms:
Staphylococci (coagulase-positive as welll as penicillinase-producing strains)
Streptococci
Pneumococci
Corynebacterium diphtheriae
Bacillus anthracis
Clostridia, Listeria monocytogenes and Bacillus subtilis, Bacteroides melaninogenicus.
Gram-negative organisms:
Escherichia coli, Salmonella, Shigella
Neisseria
Proteus mirabilis
Haemophilus influenzea (some strains) and other Brucellacea
Klebsiella spp.
Treponema pallidium and actinomycetes.
Cephalexin hardy shows any activity against Proteus, Enterobocter, Pseudomonas aeruginosa, *Mycobacterium tuberculosis, chlamydiae,* protozoa and fungi.

PHARMACOKINETICS

Oral Cephalexin is readily and almost completely absorbed (more than 90%). As both the amount absorbed and the absorption rate are largely unaffected by gastric filling, patients with sensitive stomach may take the drug with their meals. Peak serum levels are reached after no more than 60-90 minutes following oral ingestion.

Cephalosporin readily diffuse into tissues, including bone, joints, and the pericardial as well as pleural cavities. Only both 10-15% of a dose are bound to plasma proteins. Plasma half-life is approximately 50 min. Elimination is mainly renal. As almost all of the dose recovered from the urine is therapeutically active, therapeutic urinary concentrations are even reached in patients with reduced renal function. Cephalexin is washed out of organism hemodialysis and peritoneal dialysis.

INDICATIONS

Cephalexin is used to treat septicaemia, bone and joint infections, including burn wound infections and urinary tract infections caused by susceptible bacterial organisms. They are not effective in treating meningitis.

Cephalexin is used as a possible alternative to the penicillins for staphylococcus and non-enterococcal streptococcal infections including pneumonia, bone and joint infections, and bacterial endocarditis.

RECOMMENDED DOSAGE

Adults :
1 to 2 g daily given in divided doses at 6, 8 or 12 hourly intervals.

Infants and children :
25 to 100 mg per kg body-weight daily in divided doses to a maximum of 4 g daily.

A common dosage regimen is :
Children age,
5 to 12 years - 250 mg three times daily.
1 to 5 years - 125 mg three times daily.
Under 1 year - 62.5 mg to 125 mg twice daily.

ROUTE OF ADMINISTRATION

Oral.

CONTRAINDICATIONS

Patients who are known or suspected to be allergic to other cephalosporins should not be treated with cephalexin. Cephalosporins should be given with caution to penicillin-sensitive, as there is some evidence of partial cross-allergenicity between the penicillins and the cephalosporins. Patients have had severe reactions (including anaphylaxis) to both drugs.

Cephalexin is contraindicated in patients with acute porphyria.

Severe systemic infections, which require parenteral cephalosporin treatment, should not be treated orally during the acute stage.

WARNINGS AND PRECAUTIONS

1) Serious and occasionally fatal hypersensitivity reactions (including anaphylactoid and severe cutaneous adverse reactions) have been reported in patients receiving therapy with beta-lactams. Before initiating therapy with Cephalexin, careful inquiry should be made concerning previous hypersensitivity reaction to penicillins, cephalosporins, carbapenems or other beta-lactam agents. If an allergic reaction occurs, Cephalexin must be discontinued immediately and appropriate alternative therapy instituted.

2) Care is also necessary in patients with known histories of allergy.

3) Cephalexin should be administered with caution in the presence of markedly impaired renal function as it is excreted mainly by the kidneys. Careful clinical and laboratory studies should be made because the safe dosage may be lower than that usually recommended.

4) Positive direct Coombs' tests have been reported during treatment with cephalosporin antibiotics, In haematological studies, or in transfusion cross-matching procedures when antiglobulin tests are performed on the minor side, or in Coombs' testing of new-borns whose mothers have received cephalosporin antibiotics before parturition, it should be recognised that a positive Coombs' test may be due to the drug. In addition, Cephalosporins have also been reported to cause false-positive results in urine glucose tests that contain cupric sulfate solution (e.g., Benedict's reagent, Clinitest); patients with diabetes mellitus who use urinary testing methods may receive caution regarding this interference.

5) A false positive reaction for glucose in the urine may occur with Benedict's or Fehling's solutions or with copper sulphate test tablets.

6) Prolonged use of cefalexin may result in the overgrowth of non-susceptible organisms. Careful observation of the patient is essential. If super-infection occurs during therapy, appropriate measures should be taken.

7) Pseudomembranous colitis has been reported with virtually all broad-spectrum antibiotics, including macrolides, semisynthetic penicillins and cephalosporins. It is important, therefore, to consider its diagnosis in patients who develop diarrhoea in association with the use of antibiotics. Such colitis may range in severity from mild to life-threatening. Mild cases of pseudomembranous colitis usually respond to drug discontinuance alone. In moderate to severe cases, appropriate measures should be taken.

8) Acute generalized exanthematous pustulosis (AGEP) has been reported in association with cephalexin treatment. At the time of prescription patients should be advised of the signs and symptoms and monitored closely for skin reactions. If signs and symptoms suggestive of these reactions appear, cefalexin should be withdrawn immediately and an alternative treatment considered. Most of these reactions occurred most likely in the first week during treatment.

9) Following cephalexin treatment, some patients were found to show a transient slight elevation of serum glutamic-oxaloacetic transaminase (SGOT) and SGPT (also called ALT) levels.

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10) Cephalosporins may interfere with tests for urinary ketone bodies.

11) All cephalosporins, including cephalexin, may rarely cause hypothrombinemia and have the potential to cause bleeding. The mechanism is usually via the inhibition of normal gut flora and decreases in normal vitamin K synthesis, leading to coagulopathy. Cephalosporins should be used cautiously when there is a need for prolonged antibiotic therapy or other risk factors (e.g., malnutrition). Patients with a preexisting coagulopathy (e.g., vitamin K deficiency) may be at higher risk for developing bleeding complications.

DRUG INTERACTIONS

1) Probenecid causes reduced excretion of cephalexin leading to increased plasma concentrations. Cephalosporins may have an increased risk of nephrotoxicity in the presence of highly potent diuretics (ethacrynic acid, furosemide, loop diuretics) or other potentially nephrotoxic antibiotics (aminoglycosides, polymyxin, colistin, amphotericin, capreomycin, vancomycin).

2) Hypokalaemia has been described in patient taking cytotoxic drugs for leukaemia when they were given gentamicin and cephalexin.

3) As cephalosporins like cephalexin are only active against proliferating microorganisms, they should not be combined with bacteriostatic antibiotics.

4) Combined use of cephalosporins and oral anticoagulants may prolong prothrombin time.

5) A potential interaction between cephalexin and metformin may result in an accumulation of metformin and could result in fatal lactic acidosis.

6) Concomitant use of uricosuric drugs (e.g. probenecid) suppresses renal drug elimination. As a result, cephalexin plasma levels are increased and sustained for longer periods.

7) If associated with highly potent diuretics (ethacrynic acid, furosemide) or other potentially nephrotoxic antibiotics (aminoglycosides, polymyxin, colistin), cephalosprins may show higher nephrotoxicity.

8) Cephalexin may reduce the effects of oral contraceptives.

PREGNANCY AND LACTATION

Pregnancy

Although are no evidence of teratogenicity, caution should be exercised when prescribing for the pregnant patient.

Lactation

Cephalexin is excreted in human milk. Caution should be exercised when cephalexin is administered to a nursing woman.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

There are no effects on ability to drive or to operate machinery.

ADVERSE EFFECTS

Side effects of cephalexin include gastro-intestinal disturbances such as nausea, vomiting, diarrhoea and abdominal discomfort. The most common of these effects is diarrhoea, but this is rarely severe enough to warrant cessation of therapy. Dyspepsia has also occurred. Transient hepatitis and cholestatic jaundice have rarely been reported.

Allergic reactions have been reported such as rash, urticaria, angioedema and rarely erythema multiforme, Stevens-Johnson syndrome and toxic epidermal necrolysis (exanthematic necrolysis). These reactions usually subsided upon discontinuation of the drug, although in some cases supportive therapy may be necessary. Anaphylaxis and acute generalised exanthematous pustulosis (AGEP) have also been reported.

Other side effects such as genital and anal pruritus, genital candidiasis, vaginitis and vaginal discharge, dizziness, fatigue, headache, agitation, confusion, hallucinations, arthralgia, arthritis and joint disorders have been reported.

As with other cephalosporins interstitial nephritis has rarely been reported.

Eosinophilia, neutropenia, thrombocytopenia, haemolytic anaemia and slight elevations in AST and ALT have been reported.

As with other broad-spectrum antibiotics prolonged use may result in the overgrowth of non-susceptible organisms, e.g. candida. This may present a vulvo-vaginitis.

There is a possibility of development of pseudomembranous colitis and it is therefore important to consider its diagnosis in patients who develop diarrhoea while taking cephalexin. It may range in severity from mild to life threatening with mild case usually responding to cessation of therapy. Appropriate measures should be taken with moderate to severe cases.

OVERDOSE AND TREATMENT

Signs and symptoms of oral overdose may include nausea, vomiting, epigastric distress, diarrhoea and haematuria.

Treatment: Unless 5 to 10 times the normal dose of cephalexin has been ingested, gastrointestinal decontamination should not be necessary.

In the event of severe overdosage, general supportive care is recommended including close clinical and laboratory monitoring of haematological, renal and hepatic functions and coagulation status until the patient is stable. Protect the patient's airway and support ventilation and perfusion. Meticulously monitor and maintain with acceptable limits, the patients' vital signs, blood gases, serum electrolytes, etc. Absorption of drugs from the gastrointestinal tract may be decreased by giving activated charcoal, which, in many cases is more effective than emesis or lavage ; consider charcoal instead of or in addition to gastric emptying. Repeated doses of charcoal over time may hasten elimination of some drugs that have been absorbed. Safeguard the patient's airway when employing gastric emptying or charcoal.

Force diuresis, peritoneal dialysis, haemodialysis or charcoal haemoperfusion have not been established as beneficial for an overdosage of cephalexin; however, it would be extremely unlikely that one of these procedures would be indicated.

There have been reports of haematuria without impairment of renal function in children accidentally ingesting more than 3.5 g of cefalexin in a day. Treatment has been supportive (fluids) and no sequelae have been reported.

USER INSTRUCTIONS

Take this medication at regular intervals preferably before meals and complete the prescribed course as directed by the physician.

ROUTE OF ADMINISTRATION

Oral.

STORAGE CONDITIONS

Store below 30°C.
Protect from light.

SHELF LIFE

Product should not be used beyond the expiry date imprinted on the product packaging.

PRESENTATION

Pharmaniaga Cephalexin Capsule 250 mg
In boxes of 100 capsules (10 x 10's)
and 500 capsules (50 x 10's)

Pharmaniaga Cephalexin Capsule 500 mg
In boxes of 500 capsules (50 x 10's)

Date of revision: 3 February 2022

PRODUCT REGISTRATION HOLDER / MANUFACTURER:

Pharmaniaga Manufacturing Berhad (198001006232)
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